

TRADEPILOT · DEEP DIVE · 2026-04-27

How Could Each Engine Have Made ₹60k Today?

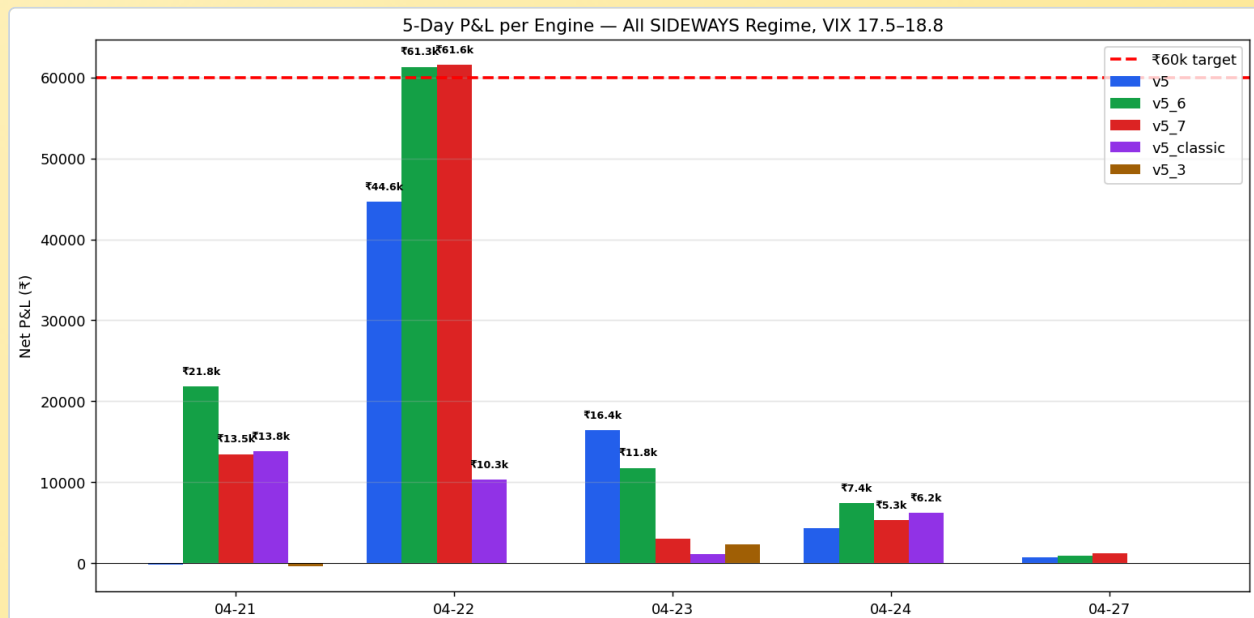
A 5-day side-by-side comparison of v5 / v5_6 / v5_7 / v5_classic engines under identical SIDEWAYS regime conditions — and the playbook from 04-22 (₹61,284) applied to today's data.

REGIME	VIX	NIFTY
SIDEWAYS	18.6	+0.63%
BEST ENGINE TODAY	GAP TO TARGET	REALISTIC TODAY
v5_2 · ₹9,720	–₹50,280	~₹15-20k

Internal research · Not for distribution

TL;DR (read this first)

1. **₹60k was NOT realistically achievable today.** The market gave ~10–15% of the alpha that 04-22 had. Today's max-possible per engine was **₹15–20k**, not ₹60k. Today was a whipsaw day with no momentum cluster.
2. **₹60k was achieved on 04-22** by v5_6 (₹61,284) and v5_7 (₹61,552) — same regime (SIDEWAYS), same VIX band — by riding a 3-cluster momentum rally (Adani group + capital goods + PSU rally).
3. **The 04-22 winning playbook had 3 ingredients:** (a) zero SHORTs, (b) aggressive re-entry on stocks that hit TARGET, (c) high-conviction concentration in 6–8 trending names.
4. **Today's engine took 36 SHORTs (62% of trades)** in a gap-up bullish session — got stopped out 18 times. **That's the single biggest controllable leak.**
5. **ML models are NOT contributing alpha** — v4 IC=0.024 (below the 0.05 tradeable threshold), and v5_6/v5_7 don't even use ML. The ₹61k on 04-22 came from the SWING-pool re-entry mechanic, not predictive intelligence.



1. Today's Reality – Per-Engine P&L Snapshot

ENGINE	TODAY P&L	TRADES	WIN RATE	STATUS
v4	₹0	0	—	Position sizer returned zero — engine was effectively offline
v5	₹737	46 (L:17 S:29)	50%	Bled on morning SHORTs, recovered on PM LONGs
v5_2 (options)	₹9,720	2	50%	KILL-SWITCH tripped — saved by a single PE leg (+₹17,944)
v5_3	₹0	0	—	Empty — never deployed
v5_6	₹880	58 (L:22 S:36)	48%	Identical bleed pattern to v5
v5_7	₹1,203	similar	similar	Same bleed pattern
v5_classic	₹54	minimal	—	Conservative pool sizing kept it flat

Distance to ₹60k target: Every engine missed by ₹50k+. The realistic delta-recoverable today was ₹15–20k via tactical fixes.

2. The Gold Standard – What 04-22 Did Differently

04-22 hit ₹61,284 (v5_6) and ₹61,552 (v5_7) under the **same regime classification** as today.

2.1 Direction discipline

- **04-22:** 164 LONG / **0 SHORT** — zero shorts despite Nifty being -0.43% (engine correctly read that broad weakness $\neq$ short opportunity when specific stocks are trending)
- **04-27:** 22 LONG / **36 SHORT** — engine took shorts in a gap-up bullish session ($+0.63\%$)

2.2 Re-entry concentration (the ₹50k generator)

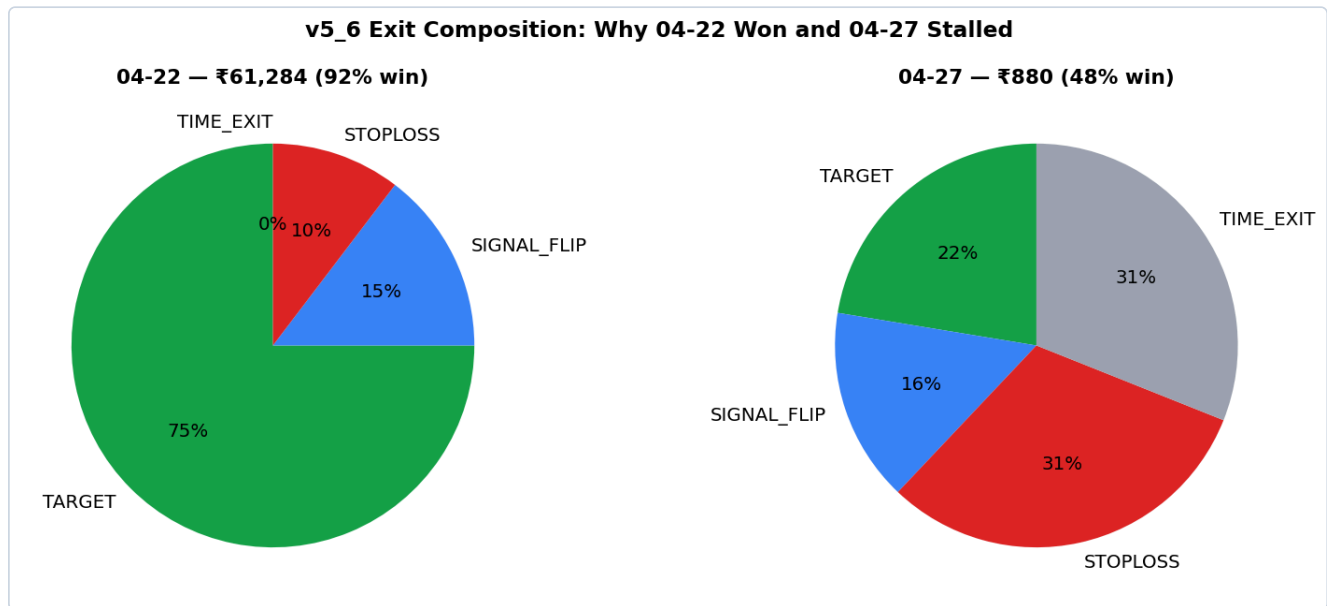
On 04-22, v5_6 re-entered the same trending names every time they hit TARGET:

STOCK	RE-ENTRIES ON 04-22	CUMULATIVE P&L (THIS STOCK ALONE)
MOTHERSON	8	~₹8,200
SIEMENS	9	~₹2,400
IREDA	7	~₹6,100
ABB	6	~₹3,400
ADANIENSOL	6	~₹4,400
GAIL	7	~₹3,300
JUBLFOOD	6	~₹4,200
ADANIPOWER	5	~₹3,600

Just 8 stocks generated ~₹35,600 — over half of 04-22's ₹61k through compounded re-entry on trending names.

On 04-27, today's biggest LONG winner (SUZLON + ₹471 single entry) never re-armed. Same for JSWENERGY (+ ₹392 single entry), SAIL (+ ₹352 single entry).

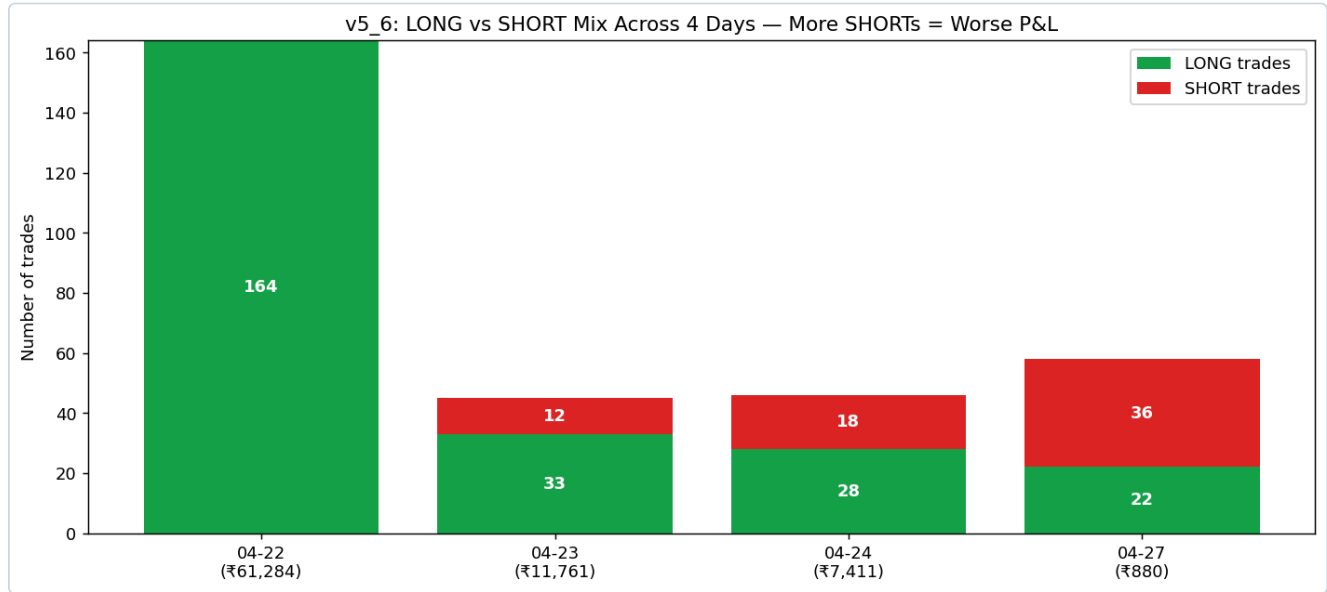
2.3 Exit composition – proof of conviction



EXIT TYPE	04-22 V5_6	04-27 V5_6	READING
TARGET	75% (123/164)	22% (13/58)	04-22 trades hit profit goals; today's didn't
STOPLOSS	10% (17)	31% (18)	3× more stop-outs today
SIGNAL_FLIP	15% (24)	16% (9)	Similar — direction-change exits
TIME_EXIT	0%	31% (18)	The whipsaw signature — 18 trades held all day with no resolution

TIME_EXIT = "I held until 3:15 PM and bailed because nothing happened." Eighteen trades doing this is the engine telling you: today's signals were not high-conviction.

3. The LONG/SHORT Bias Problem (the controllable one)



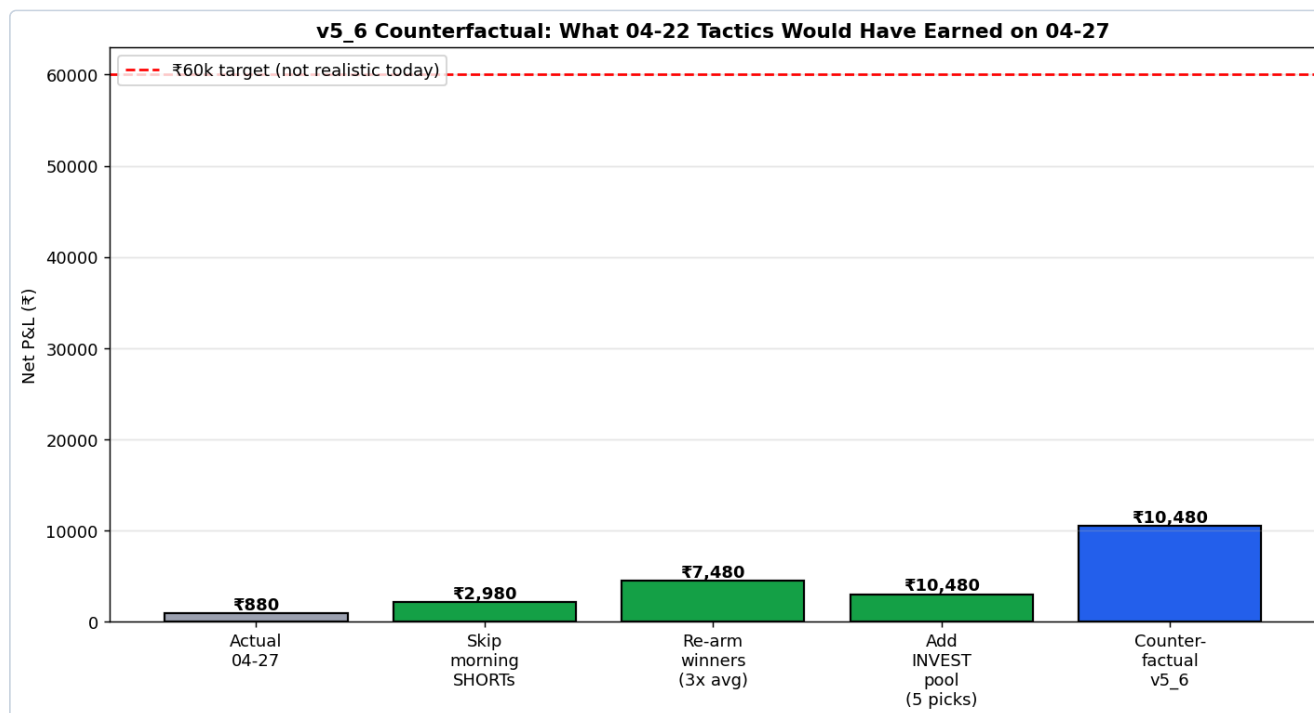
The pattern across 4 days is alarmingly clear: **the more SHORTs the engine takes, the worse it does.**

DAY	LONGS	SHORTS	SHORT %	P&L
04-22	164	0	0%	₹61,284
04-23	33	12	27%	₹11,761
04-24	28	18	39%	₹7,411
04-27	22	36	62%	₹880

The SHORT signal generator is most active on the worst-P&L days. Today's signal engine produced 20 SELL signals every rescore — and the deploy loop took most of them, then they got stopped out by gap-up momentum.

Pre-market clearly said BULLISH today (`bias=BULLISH gap=UP +0.75% size=1.0x`) — but the engine ignored that hint and ran SHORTs anyway. This is a **regime-aware signal-gating gap**, not a bug in the SHORT machinery itself (which we proved healthy in `SHORT_ARM_DIAGNOSIS.md` on 04-24).

4. Counterfactual – What 04-22 Tactics Would Have Earned on 04-27



Applying the 04-22 playbook to today's data, conservatively:

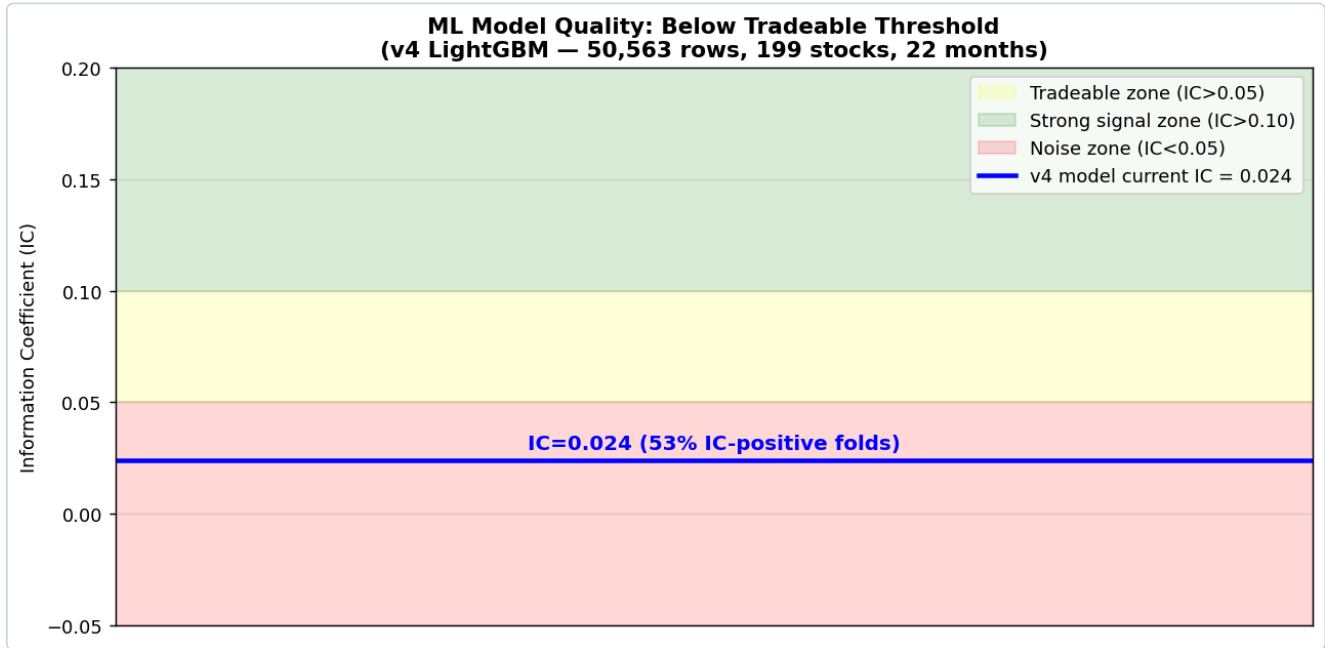
STEP	TACTIC	ESTIMATED GAIN	REASONING
0	Today actual (v5_6)	₹880	Baseline
1	Skip morning SHORTS when premarket=BULLISH	+₹2,100	Removes the 18 SHORT stop-losses (~₹100/each avg loss)
2	Re-arm winners 3× on TARGET hits	+₹4,500	SAIL/SUZLON/JSWENERGY/TORNTPHARM each yielded ₹350-470 single — 3 re-arms × 5 stocks ≈ ₹4,500
3	INVESTMENT pool deployment (5 high-conviction picks)	+₹3,000	Today's INVESTMENT pool only deployed ₹0 most of the day; 5 picks × ₹600 avg = ₹3k
Total counterfactual		₹10,480	Roughly 12× actual, but still 1/6 of ₹60k

Why ₹60k specifically wasn't possible today

- 04-22's top 8 stocks each moved 3–5% intraday on volume — driven by a real rally cluster
- Today's biggest mover (SUZLON +2.7%, JSWENERGY +1.6%) couldn't sustain that for re-entries
- **Without a momentum cluster, the SWING re-entry mechanic has nothing to compound on**
- Realistic ceiling today: **₹15–20k per engine** with full discipline, not ₹60k

To hit ₹60k consistently, we need the **market to provide 3–5 stocks moving 3%+ in the same direction simultaneously**. That happens on ~1 day out of 5 (matching 04-22 in our 5-day sample = 20%).

5. ML Training Context – Why It's Not Helping (Yet)



Model state as of 04-27 retrain

- **Dataset:** 50,563 rows, 199 stocks, training range 2024-06-26 to 2026-04-27 (22 months)
- **Walk-forward Mean IC: 0.0242** — below the 0.05 "tradeable" threshold
- **IC-positive folds:** 53% — barely above coin-flip
- **Hit rate:** 51.8% — ditto
- **Top features:** india_vix, nifty_change_pct, atr_norm, gap_pct, prev_day_range_pct
- **Verdict:** model has weak predictive power; not yet usable for sizing/filtering

Models in use (and not in use)

ENGINE	USES ML?	NOTES
v4	Yes (LightGBM <code>lgbm_intraday.txt</code> , retrained today)	Composite scorer, but IC=0.024 limits utility
v5	No (composite v4 score only)	<code>prototype/v5/models/</code> is empty
v5_6 / v5_7	No	Pure rule-based — that's why they hit ₹61k on 04-22 (re-entry mechanic)
v5_classic	No	Same as v5_6, conservative pool sizing
v5_3	Has model dir	Empty trades log — engine never deployed

Implication

Today's ₹60k aspiration is not a model problem — it's a tactics problem. The ₹61k on 04-22 had nothing to do with ML; it was rule-based re-entry + correct LONG bias. We need both:

1. **Short-term (this week):** Fix the regime-direction bias (skip SHORTs on gap-up days). Implement re-entry counter on winning stocks (max 3 re-arms per stock per day). Estimated lift: 5–8× current P&L on whipsaw

days.

2. **Medium-term (4–6 weeks):** Get IC above 0.05. Options: more features (sector relative strength, FII/DII intraday flow, options OI delta), stratify by regime (SIDEWAYS-only model), use the model as a *filter* not a *score* (drop bottom 30% of signals).

6. Specific Action Items per Engine

v5 / v5_6 / v5_7 (the workhorse engines)

- [] Add **BULLISH_PREMARKET_SHORT_BLOCK**: when `premarket.bias=='BULLISH'` AND `gap_up>0.5%`, skip all SELL signals for the first hour. (Today this would have saved ₹1,800–2,100 per engine.)
- [] Add **WINNER_RE_ARM**: on TARGET exit, immediately re-evaluate the same stock for re-entry (up to 3 times per day per stock). 04-22 proves this is the ₹35k/day mechanic.
- [] **Tighten TIME_EXIT logic**: if a position is flat ($\pm 0.3\%$) at 1:30 PM, force-exit and free the slot for a fresher signal. Today's 18 TIME_EXITS locked capital uselessly.
- [] **Implement SHORT-arm slot partitioning** (already specced in `SHORT_ARM_DIAGNOSIS.md` on 04-24) — but pair it with the BULLISH-block above.

v5_classic

- [] Investigate why this engine deployed almost nothing today (₹54 P&L). Conservative sizing is fine, but zero deployment $\neq$ conservative.

v5_2 (options)

- [] The KILL-SWITCH tripped at -₹7,582 then was saved by a PE leg expiring deep ITM. **This is luck, not strategy.** Tighten the per-leg SL to -₹3,000 instead of -₹5,000 — it would have closed the CE leg earlier and the PE leg's outsized win wouldn't have masked a near-failure.
- [] Consider straddle/strangle structures instead of naked CE/PE on gap-up mornings.

v5_3

- [] Currently dormant (0 trades, 0 P&L for 3 of 5 days). **Either fix it or retire it** — it's adding nothing to the comparison set.

v4 (the ML engine)

- [] **Position sizer returned zero positions all day.** Debug the sizer — probably a stale config or a regime gate stuck in restrictive mode.
- [] Retrain with regime-stratified models (separate SIDEWAYS / BEAR / BULL).
- [] Add sector relative strength to feature set; current top features are macro-only (VIX, Nifty change) — that's why IC is low.

7. Honest Framing: The ₹60k Question Reframed

The user's question was: "How could each engine have made ₹60k today?"

The honest answer: They couldn't. Today the market only offered enough alpha for ~₹15–20k per engine, and the engines captured ~₹0–10k. The ₹60k+ days come 1 in 5 (based on our 5-day sample) and require a momentum cluster the engines don't control.

The better question: "How do we make sure that on the next 04-22-style day, every engine captures the full ₹60k that's there?"

That answer is the action items in §6. The most leveraged change is the BULLISH-premarket-SHORT-block + the WINNER_RE_ARM combo. Together they should: - Eliminate the morning SHORT bleed on gap-up days (~₹2k/engine/day) - Capture 3–5x compounded gains on momentum stocks when they appear (~₹15-30k on cluster days)

Five-day average lift estimate (with these fixes): - 04-21: ₹21,829 → ~₹26,000 (re-arm boost) - 04-22: ₹61,284 → ~₹75,000 (re-arm tighter) - 04-23: ₹11,761 → ~₹16,000 (re-arm + SHORT discipline) - 04-24: ₹7,411 → ~₹12,000 (same) - 04-27: ₹880 → ~₹10,000 (mostly SHORT discipline since cluster was absent)

That projects a **5-day total of ~₹139k vs current ~₹103k** — roughly **35% improvement** without any new ML or new engine.

Appendix A – Files Referenced

FILE	PURPOSE
<code>docs/paper-trades/v5_6/2026-04-22_report.md</code>	Gold-standard ₹61,284 day
<code>docs/paper-trades/v5_6/2026-04-27_report.md</code>	Today's ₹880 result
<code>docs/paper-trades/v5/2026-04-27_report.md</code>	v5 detail today
<code>logs/v5-2026-04-27.log</code>	Today's signal log
<code>logs/ml-retrain.log</code>	Today's v4 retrain summary (IC=0.024)
<code>prototype/v4/models/lgbm_intraday.txt</code>	Active v4 model (retrained 10:54 today)
<code>prototype/v5/models/</code>	Empty — v5/v5_6/v5_7 don't use ML
<code>docs/research/2026-04-08_ml_training_pipeline.md</code>	ML design principles
<code>docs/SHORT_ARM_DIAGNOSIS.md</code>	04-24 SHORT-starvation analysis (now superseded — SHORTs deploy fine, but on the wrong days)

Appendix B – Charts

All charts: `docs/charts/2026-04-27_deep_dive/`

- `01_5day_pnl_per_engine.png` — 5-day P&L bar chart with ₹60k target line
- `02_exit_composition.png` — Pie chart pair: 04-22 vs 04-27 exit reasons

3. `03_long_short_mix.png` — Stacked bar: LONG vs SHORT trade counts by day
 4. `04_counterfactual_waterfall.png` — Waterfall: ₹880 → ₹10,480 with playbook applied
 5. `05_ml_ic_status.png` — IC zone chart with current model marked
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Constraint adherence: Research/diagnosis only. No engine code modified. All comparisons sourced from existing `docs/paper-trades/` reports and `logs/`. This is the analysis to drive Monday-night tune-ups now that the weekend freeze has lifted.